Describing Data: Frequency Tables, Frequency Distributions, and Graphic Presentation

Chapter 2

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Learning Objectives

LO2-1 Summarize qualitative variables with frequency and relative frequency tables

LO2-2 Display a frequency table using a bar or pie chart

LO2-3 Summarize quantitative variables with frequency and relative frequency distributions

LO2-4 Display a frequency distribution using a histogram or frequency polygon

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Constructing Frequency Tables

FREQUENCY TABLE A grouping of qualitative data into mutually exclusive and collectively exhaustive classes showing the number of observations in each class.

- Mutually exclusive means the data fit in just one class
- Collectively exhaustive means there is a class for each value

TABLE 2–1 Frequency Table for Vehicles Sold Last Month at Applewood Auto Group by Location

Location	Number of Cars
Kane	52
Olean	40
Sheffield	45
Tionesta	43
Total	180

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Constructing Frequency Tables (2 of 3)

- To construct a frequency table
  - First sort the data into classes
  - Count the number in each class and report as the class frequency

TABLE 2–1 Frequency Table for Vehicles Sold Last Month at Applewood Auto Group by Location

Location	Number of Cars
Kane	52
Olean	40
Sheffield	45
Tionesta	43
Total	180

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Constructing Frequency Tables (3 of 3)

- Convert each frequency to a relative frequency
  - Each of the class frequencies is divided by the total number of observations
  - Shows the fraction of the total number observations in each class

TABLE 2–2 Relative Frequency Table of Vehicles Sold by Location Last Month at Applewood Auto Group

Location	Number of Cars	Relative Frequency	Found by
Kane	52	.289	52/180
Olean	40	.222	40/180
Sheffield	45	.250	45/180
Tionesta	43	.239	43/180
Total	180	1.000	

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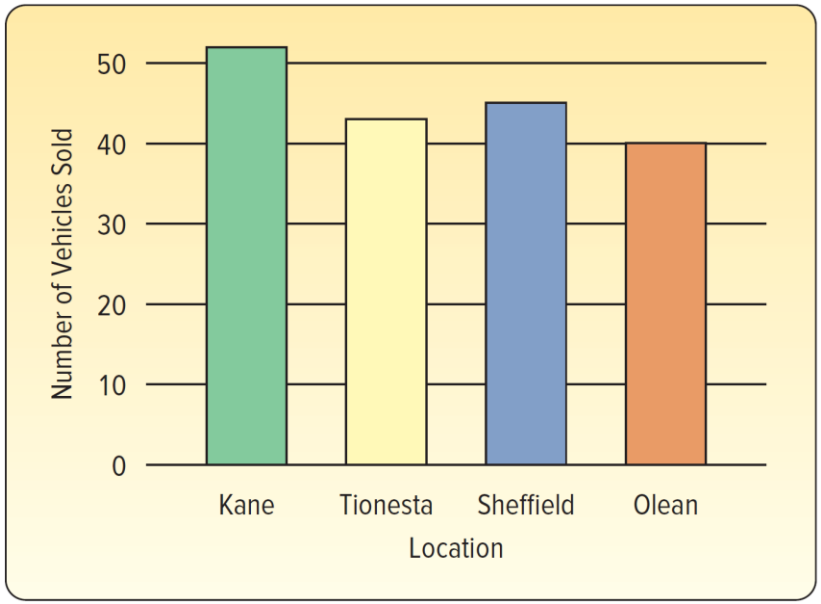
Graphic Presentation of Qualitative Data

BAR CHART A graph that shows the qualitative classes on the horizontal axis and the class frequencies on the vertical axis. The class frequencies are proportional to the heights of the bars.

- Use a bar chart when you wish to compare the number of observations for each class of a qualitative variable.



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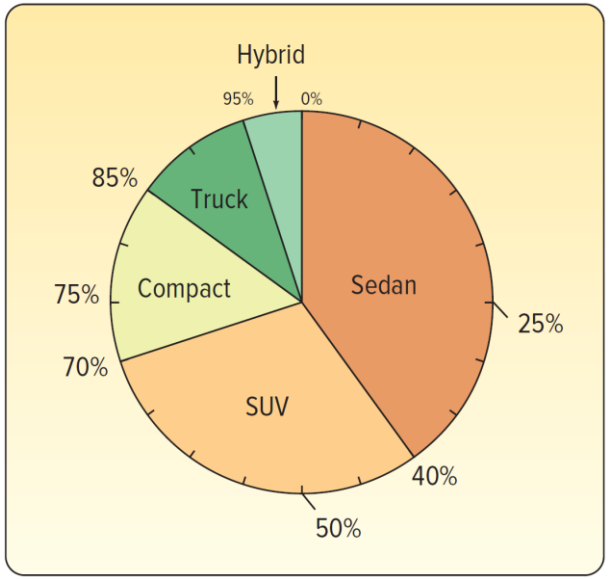
Graphic Presentation of Qualitative Data (2 of 2)

PIE CHART A chart that shows the proportion or percentage that each class represents of the total number of frequencies.

- Use a pie chart when you wish to compare relative differences in the percentage of observations for each class of a qualitative variable.



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Constructing Frequency Distributions

FREQUENCY DISTRIBUTION A grouping of quantitative data into mutually exclusive and collectively exhaustive classes showing the number of observations in each class.

- This is a four-step process

1. Decide on the number of classes
2. Determine the class interval
3. Set the individual class limits
4. Tally the data into classes and determine the number of the observations in each class

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Frequency Distributions

- Step 1 Decide on the number of classes
- Use the $2^k > n$ rule, where n=180
  - k is the number of classes
  - n is the number of values in the data set
  - $2^k > 180$, let k = 8
  - So use 8 classes

TABLE 2–4 Profit on Vehicles Sold Last Month by the Applewood Auto Group

Maximum

$1,387	$2,148	$2,201	$ 963	$ 820	$2,230	$3,043	$2,584	$2,370
1,754	2,207	996	1,298	1,266	2,341	1,059	2,666	2,637
1,817	2,252	2,813	1,410	1,741	3,292	1,674	2,991	1,426
1,040	1,428	323	1,553	1,772	1,108	1,807	934	2,944
1,273	1,889	352	1,648	1,932	1,295	2,056	2,063	2,147
1,529	1,166	482	2,071	2,350	1,344	2,236	2,083	1,973
3,082	1,320	1,144	2,116	2,422	1,906	2,928	2,856	2,502
1,951	2,265	1,485	1,500	2,446	1,952	1,269	2,989	783
2,692	1,323	1,509	1,549	369	2,070	1,717	910	1,538
1,206	1,760	1,638	2,348	978	2,454	1,797	1,536	2,339
1,342	1,919	1,961	2,498	1,238	1,606	1,955	1,957	2,700
443	2,357	2,127	294	1,818	1,680	2,199	2,240	2,222
754	2,866	2,430	1,115	1,824	1,827	2,482	2,695	2,597
1,621	732	1,704	1,124	1,907	1,915	2,701	1,325	2,742
870	1,464	1,876	1,532	1,938	2,084	3,210	2,250	1,837
1,174	1,626	2,010	1,688	1,940	2,639	377	2,279	2,842
1,412	1,762	2,165	1,822	2,197	842	1,220	2,626	2,434
1,809	1,915	2,231	1,897	2,646	1,963	1,401	1,501	1,640
2,415	2,119	2,389	2,445	1,461	2,059	2,175	1,752	1,821
1,546	1,766	335	2,886	1,731	2,338	1,118	2,058	2,487

Minimum

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Frequency Distributions (2 of 4)

- Step 2 Determine the class interval, i

$$i \geq \frac{\textit{Maximum Value} - \textit{Minimum Value}}{k}$$

  - Round up to some convenient number

$$i \geq \frac{\$3{,}292 - \$294}{8} = \$374.75$$

  - So decide to use an interval of $400
  - The interval is also referred to as the class width

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Frequency Distributions (3 of 4)

- Step 3 Set the individual class limits
  - Lower limits should be rounded to an easy-to-read number when possible

Classes
$ 200 up to $ 600
600 up to 1,000
1,000 up to 1,400
1,400 up to 1,800
1,800 up to 2,200
2,200 up to 2,600
2,600 up to 3,000
3,000 up to 3,400

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Frequency Distributions (4 of 4)

- Step 4 Tally the individual data into the classes and determine the number of observations in each class
  - The number of observations is the class frequency

Profit	Frequency
$ 200 up to $ 600	卌 III
600 up to 1,000	卌 卌 I
1,000 up to 1,400	卌 卌 卌 卌 III
1,400 up to 1,800	卌 卌 卌 卌 卌 卌 卌 III
1,800 up to 2,200	卌 卌 卌 卌 卌 卌 卌 卌 卌
2,200 up to 2,600	卌 卌 卌 卌 卌 II
2,600 up to 3,000	卌 卌 卌 IIII
3,000 up to 3,400	IIII

Profit	Frequency
$ 200 up to $ 600	8
600 up to 1,000	11
1,000 up to 1,400	23
1,400 up to 1,800	38
1,800 up to 2,200	45
2,200 up to 2,600	32
2,600 up to 3,000	19
3,000 up to 3,400	4
Total	180

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Relative Frequency Distributions

- To find the relative frequencies, simply take the class frequency and divide by the total number of observations

Profit	Frequency	Relative Frequency	Found by
$ 200 up to $ 600	8	.044	8/180
600 up to 1,000	11	.061	11/180
1,000 up to 1,400	23	.128	23/180
1,400 up to 1,800	38	.211	38/180
1,800 up to 2,200	45	.250	45/180
2,200 up to 2,600	32	.178	32/180
2,600 up to 3,000	19	.106	19/180
3,000 up to 3,400	4	.022	4/180
Total	180	1.000	

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Graphic Presentation of a Frequency Distribution

HISTOGRAM A graph in which the classes are marked on the horizontal axis and the class frequencies on the vertical axis. The class frequencies are represented by the heights of the bars, and the bars are drawn adjacent to each other.

- A histogram shows the shape of a distribution
- Each class is depicted as a rectangle, with the height of the bar representing the number in each class

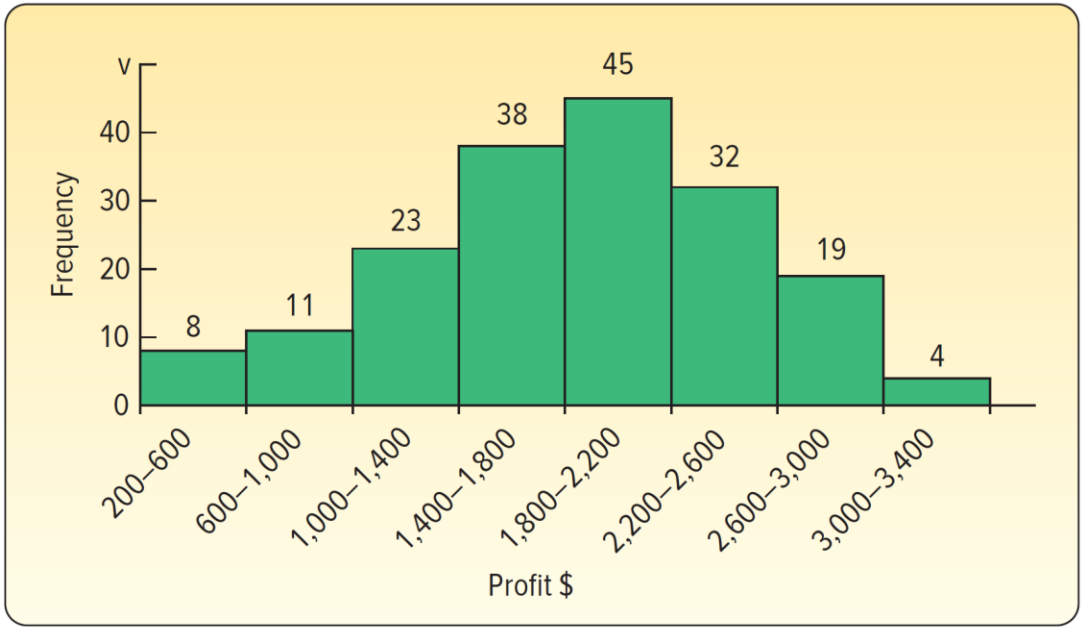

CHART 2–4 Histogram of the Profit on 180 Vehicles Sold at the Applewood Auto Group

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Graphical Presentation of a Frequency Distribution

- A frequency polygon, similar to a histogram, also shows the shape of a distribution
- These are good to use when comparing two or more distributions

Profit	Midpoint	Frequency
$ 200 up to $ 600	$ 400	8
600 up to 1,000	800	11
1,000 up to 1,400	1,200	23
1,400 up to 1,800	1,600	38
1,800 up to 2,200	2,000	45
2,200 up to 2,600	2,400	32
2,600 up to 3,000	2,800	19
3,000 up to 3,400	3,200	4
Total		180

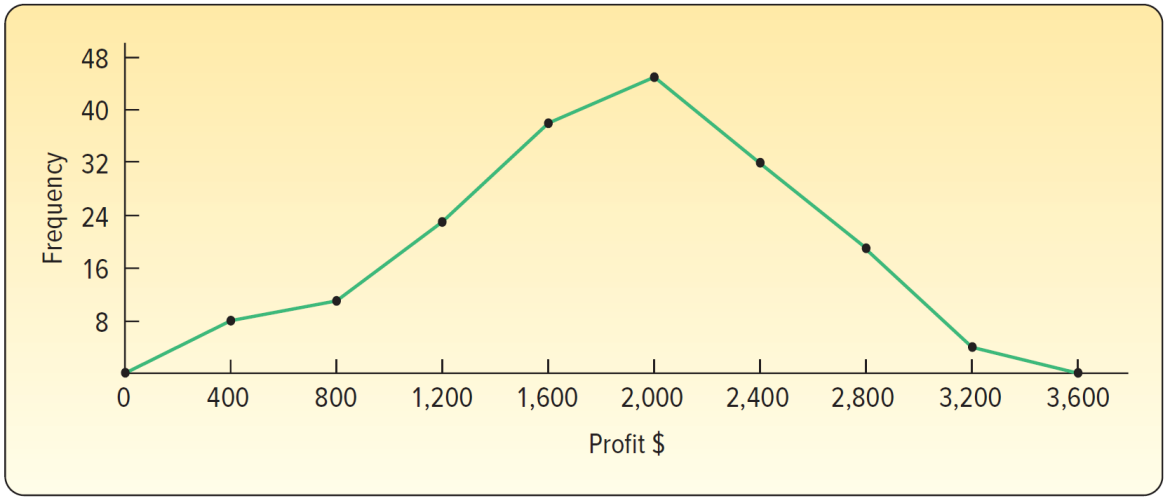

CHART 2–5 Frequency Polygon of Profit on 180 Vehicles Sold at Applewood Auto Group

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Cumulative Frequency Distributions

- To construct a cumulative frequency distribution, add each frequency to the frequencies before it
- This shows how many values have accumulated as you move from one class down to the next class

TABLE 2–8 Cumulative Frequency Distribution for Profit on Vehicles Sold Last Month at Applewood Auto Group

Profit	Cumulative Frequency	Found by
Less than $ 600	8	8
Less than 1,000	19	8 + 11
Less than 1,400	42	8 + 11 + 23
Less than 1,800	80	8 + 11 + 23 + 38
Less than 2,200	125	8 + 11 + 23 + 38 + 45
Less than 2,600	157	8 + 11 + 23 + 38 + 45 + 32
Less than 3,000	176	8 + 11 + 23 + 38 + 45 + 32 + 19
Less than 3,400	180	8 + 11 + 23 + 38 + 45 + 32 + 19 + 4

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Cumulative Relative Frequency Distribution

- To construct a cumulative relative frequency distribution, we divide the cumulative frequencies by the total number of observations

TABLE 2–9 Cumulative Relative Frequency Distribution for Profit on Vehicles Sold Last Month at Applewood Auto Group

Profit	Cumulative Frequency	Cumulative Relative Frequency
Less than $ 600	8	8/180 = 0.044 = 4.4%
Less than 1,000	19	19/180 = 0.106 = 10.6%
Less than 1,400	42	42/180 = 0.233 = 23.3%
Less than 1,800	80	80/180 = 0.444 = 44.4%
Less than 2,200	125	125/180 = 0.694 = 69.4%
Less than 2,600	157	157/180 = 0.872 = 87.2%
Less than 3,000	176	176/180 = 0.978 = 97.8%
Less than 3,400	180	180/180 = 1.000 = 100%

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Cumulative Frequency Polygon

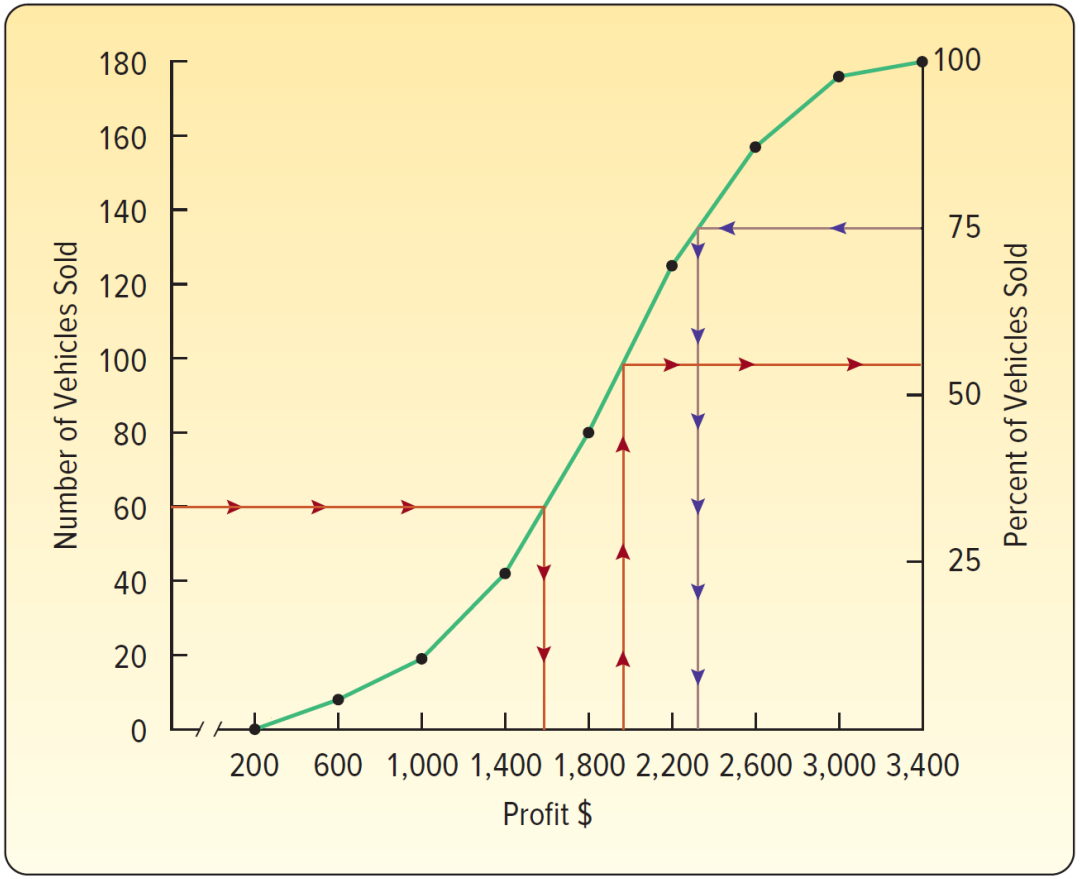

CHART 2–7 Cumulative Frequency Polygon for Profit on Vehicles Sold Last Month at Applewood Auto Group

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Chapter 2 Practice Problems

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Question 5

LO2-2

Wellstone Inc. produces and markets replacement covers for cell phones in five different colors: bright white, metallic black, magnetic lime, tangerine orange, and fusion red. To estimate the demand for each color, the company set up a kiosk for several hours in the Mall of America and asked randomly selected people which cover color was their favorite.

Bright white	130
Metallic black	104
Magnetic lime	325
Tangerine orange	455
Fusion red	286

a. What is the table called?
b. Draw a bar chart for the table.
c. Draw a pie chart.
d. If Wellstone Inc. plans to produce one million cell phone covers, how many of each color should it produce?

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Question 11

LO2-3

Wachesaw Manufacturing Inc. produced the following number of units in the last 16 days.

27	27	27	28	27	25	25	28
26	28	26	28	31	30	26	26

The information is to be organized into a frequency distribution.

a. How many classes would you recommend?

b. What class interval would you suggest?

c. What lower limit would you recommend for the first class?

d. Organize the information into a frequency distribution and determine the relative frequency distribution.

e. Comment on the shape of the distribution.

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Question 17

LO2-4

The following frequency distribution reports the number of frequent flier miles, reported in thousands, for employees of Brumley Statistical Consulting Inc. during the most recent quarter.

Frequent Flier Miles (000)	Number of Employees
0 up to 3	5
3 up to 6	12
6 up to 9	23
9 up to 12	8
12 up to 15	2
Total	50

a. How many employees were studied?
b. What is the midpoint of the first class? What lower limit would you recommend for the first class?
c. Construct a histogram.
d. A frequency polygon is to be drawn. What are the coordinates of the plot for the first class?
e. Construct a frequency polygon.
f. Interpret the frequent flier miles accumulated using the two charts.

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